Brian K. Kirui

Postdoctoral Researcher · Epidemiology

University of Gothenburg, Gothenburg, Sweden

+46 | brian.kirui@gu.se | briankibiwott.github.io | BrianKibiwott | briankkirui

Education

University of Gothenburg — *Gothenburg, Sweden*
Ph.D. Epidemiology — *2022–2026*

University of Gothenburg — *Gothenburg, Sweden*
Master of Medical Science (MMSc) in Global Health — *2019–2021*

Pharmacy and Poisons Board — *Nairobi, Kenya*
Licensed and Registered Pharmacist (RPh), Kenya — *2018*

Kenyatta University — *Nairobi, Kenya*
Bachelor of Pharmacy (BPharm) — *2011–2016*

Research Experience

Department of Public Health and Clinical Medicine, The OLIN and Sunderby Research Unit, Umeå University — *Umeå, Sweden*
Postdoctoral Researcher, Obstructive Lung Diseases — *2026 – Present*

- Conduct epidemiological research on obstructive lung diseases using population-based cohorts and healthcare registers.
- Develop and lead studies investigating disease progression, multimorbidity, and long-term health outcomes.
- Collaborate with national and international research groups on respiratory and public health research initiatives.

School of Public Health and Community Medicine, University of Gothenburg — *Gothenburg, Sweden*
Doctoral Student, Epidemiology — *May 2022 – Present*

- Investigating risk factors and long-term outcomes of COVID-19 among individuals with obstructive lung disease using Swedish population registers.
- Conduct epidemiological and register-based studies utilizing nationwide healthcare and demographic databases.
- Perform advanced statistical analyses and contribute to scientific publications in respiratory and public health research.

School of Public Health and Community Medicine, University of Gothenburg — *Gothenburg, Sweden*
Associate Researcher — *Aug 2021 – Feb 2022*

- Supported studies examining COVID-19 outcomes among patients with asthma and chronic obstructive pulmonary disease.
- Contributed to study design, statistical analysis plans, implementation, reporting, and research documentation.

School of Public Health and Community Medicine, University of Gothenburg — *Gothenburg, Sweden*
Master's Thesis Researcher — *Jan 2021 – Jun 2021*

- Conducted register-based research on epidemiological risk factors for COVID-19 among individuals with asthma and COPD in Sweden.
- Applied epidemiological and statistical methods using Swedish population-based health registers.

School of Public Health and Community Medicine, University of Gothenburg — *Gothenburg, Sweden*
Research Intern — *Nov 2020 – Jan 2021*

- Completed advanced training in Swedish register-based research and statistical analysis.
- Described characteristics of individuals with pre-existing respiratory conditions who contracted COVID-19 using national register data.

Publications

** Co-first author.*

Peer-Reviewed Publications

1. **Kirui BK**, Vikjord SA, Li H, Vanfleteren LEGW, Nyberg F. Reproducible smoker's paradox in COVID-19: comparative evidence from Swedish and Norwegian cohorts. *Scientific Reports*. 2026. In press. doi:10.1038/s41598-026-71946-9.

2. **Kirui BK**, Li H, Wallström O, Vanfleteren LEGW, Santosa A, Bonander C, Stridsman C, Börjesson M, Nyberg F. Cardiovascular and mortality risks following COVID-19-related vs. non-COVID-19 COPD exacerbations. *Annals of the American Thoracic Society*. 2026. doi:10.1093/annalsats/aaoag084. Epub ahead of print.

3. Li H, Xu Y, **Kirui B**, Santosa A, Nyberg F. Ursodeoxycholic acid treatment did not show protective effect for severe COVID-19 outcomes: a nationwide register study. *BMC Public Health*. 2026;26(1):1240. doi:10.1186/s12889-026-26908-1.

4. **Kirui BK**, Santosa A, Li H, Vanfleteren LEGW, Stridsman C, Nyberg F. Key characteristics of asthma patients with COVID-19 vary substantially by age. *Journal of Asthma and Allergy*. 2024;17:589–600. doi:10.2147/JAA.S456145.

5. Santosa A, Oras J, Li H, Nwaru C, **Kirui B**, Nyberg F. Survival of critically ill COVID-19 patients in Sweden during the first two and a half years of the pandemic. *Critical Care Medicine*. 2024;52(8):1194–1205. doi:10.1097/CCM.0000000000006271.

6. Einarsdottir MJ[*], **Kirui BK**[*], Li H, Olsson D, Johannsson G, Nyberg F, Ragnarsson O. Impact of chronic oral glucocorticoid treatment on mortality in patients with COVID-19: analysis of a population-based cohort. *BMJ Open*. 2024;14(3):e080640. doi:10.1136/bmjopen-2023-080640.

7. Labor M, **Kirui BK**, Nyberg F, Vanfleteren LEGW. Regular inhaled corticosteroids use may protect against severe COVID-19 outcome in COPD. *International Journal of Chronic Obstructive Pulmonary Disease*. 2023;18:1701–1712. doi:10.2147/COPD.S404913.

8. **Kirui BK**, Santosa A, Vanfleteren LEGW, Li H, Franzén S, Stridsman C, Nyberg F. Pre- and post-vaccination characteristics and risk factors for COVID-19 outcomes in a Swedish population-based cohort of COPD patients. *ERJ Open Research*. 2023;9(3):00711-2022. doi:10.1183/23120541.00711-2022.

9. Xu Y, Li H, **Kirui B**, Santosa A, Gisslén M, Leach S, Wettermark B, Vanfleteren LEGW, Nyberg F. Effectiveness of COVID-19 vaccines over 13 months covering the period of the emergence of the Omicron variant in the Swedish population. *Vaccines*. 2022;10(12):2074. doi:10.3390/vaccines10122074.

10. Nyberg F, Lindh M, Vanfleteren LEGW, Hammar N, Wettermark B, Sundström J, Santosa A, **Kirui BK**, Gisslén M. Adverse events of special interest for COVID-19 vaccines: background incidences vary by sex, age and time period and are affected by the pandemic. *medRxiv*. 2021. doi:10.1101/2021.10.04.21263507.

11. Jang SY, Hussain-Alkhateeb L, Rivera Ramirez T, Al-Aghbari AA, Chackalackal DJ, Cardenas-Sanchez R, Carrillo MA, Oh IH, Alfonso-Sierra EA, Oechsner P, **Kirui B**, Anto M, Diaz-Monsalve S, Kroeger A. Factors shaping the COVID-19 epidemic curve: a multi-country analysis. *BMC Infectious Diseases*. 2021;21(1):1032. doi:10.1186/s12879-021-06714-3.

Awards, Fellowships, & Grants

2026	**Swedish Heart-Lung Foundation Travel Grant**, Swedish Heart-Lung Foundation	*SEK 12,000*
2025	**Swedish Heart-Lung Foundation Travel Grant**, Swedish Heart-Lung Foundation	*SEK 24,000*
2025	**Wilhelm & Martina Lundgren Science Fund Grant**, Stiftelsen Wilhelm och Martina Lundgrens Vetenskapsfond	*SEK 25,000*
2024	**Swedish Heart-Lung Foundation Travel Grant**, Swedish Heart-Lung Foundation	*SEK 12,000*
2024	**Sahlgrenska Academy Travel Grant, Iwan & Eleonore Ljunggrens Fond**, Sahlgrenska Academy, University of Gothenburg	*SEK 8,000*
2023	**Adlerbertska Foundation Travel Grant**, Institute of Medicine, Sahlgrenska Academy, University of Gothenburg	*SEK 20,000*
2023	**Swedish Heart-Lung Foundation Travel Grant**, Swedish Heart-Lung Foundation	*SEK 20,000*
2023	**Sahlgrenska Academy Travel Grant, Anna Cederbergs Fond**, Sahlgrenska Academy, University of Gothenburg	*SEK 10,000*
2023	**FORTE EpiDem Network Junior Conference Sponsored Attendance**, FORTE EpiDem Network	

2022	**Adlerbertska Foundation Travel Grant**, Institute of Medicine, Sahlgrenska Academy, University of Gothenburg	*SEK 20,000*
2020	**Adlerbertska Foundation Scholarship**, Institute of Medicine, Sahlgrenska Academy, University of Gothenburg	*SEK 15,000*

Presentations

Invited Talks & Research Visits

Mar 2025	**Visiting Analyst: Smoking and COVID-19 in the HUNT Data**, Trondheim University Hospital & NTNU, Trondheim, Norway
Jun 2023	**Optimizing Cox Regression: Comparing Traditional Covariate Adjustment and Modern Propensity Score Methods for Balancing Covariates**, Department of Endocrinology, Sahlgrenska University Hospital
Oct 2022	**Pharmacoepidemiology and Drug Utilization Research**, 17th Hospital Pharmacists Association of Kenya Conference (Virtual)

Selected Conference Presentations

2026. *Why background incidence matters: Sex-, temporal-, and pandemic-related variation in adverse events of special interest shape COVID-19 vaccine safety signals*. Poster presentation, 42nd International Conference on Pharmacoepidemiology & Therapeutic Risk Management (ICPE), Milan, Italy. 29 August–3 September 2026.

2026. *Reproducible smoker's paradox in COVID-19: Comparative evidence from Swedish and Norwegian cohorts*. Poster presentation, Nordic Conference of Epidemiology and Register-Based Health Research (NordicEpi) and Nordic Pharmacoepidemiology (NorPEN) Conference, Tromsø, Norway. 3–5 June 2026.

2025. *Understanding age-specific risk factors for COVID-19: A machine learning approach*. Poster presentation, 41st International Conference on Pharmacoepidemiology & Therapeutic Risk Management (ICPE), Washington, DC, USA. 22–26 August 2025.

2024. *COVID-19 Risk and Effect Modification by Vaccination in Asthma*. Oral presentation, European Respiratory Society (ERS) Congress 2024, Vienna, Austria.

2024. *Risk of Acute Cardiovascular Events in COPD Patients After Experiencing COVID-19-Related vs. Non-COVID-19-Related Exacerbations: A Population-Based Study*. Oral presentation, 40th International Conference on Pharmacoepidemiology & Therapeutic Risk Management (ICPE), Berlin, Germany.

2023. *Smoking Status and COVID-19 Related Risk in Four Swedish Non-Communicable Disease Registers: A Non-Intuitive Association?* Oral presentation, Swedish Register-Based Research Summit, Stockholm, Sweden.

2023. *Impact of Chronic Oral Glucocorticoid Treatment on Mortality in Patients with COVID-19: An Analysis from a Population-Based Cohort*. Poster presentation, 39th International Conference on Pharmacoepidemiology & Therapeutic Risk Management (ICPE), Halifax, Canada.

2023. *The Role of Smoking Status as a Risk Factor in COVID-19 Infection and Severity in a Swedish Population-Based Cohort*. Poster presentation, Swedish Lung Congress 2023, Gothenburg, Sweden.

2022. *Inhaled Corticosteroids and COVID-19 Risk and Outcomes: Findings from a Population-Based Cohort Within the SCIFI-PEARL Project*. Oral presentation, ICPE 2022, Copenhagen, Denmark.

2022. *Characteristics of Asthma Patients with COVID-19 by Disease Severity and Across Age Groups: A Swedish Population-Based Analysis Within the SCIFI-PEARL Project*. Poster presentation, ICPE 2022, Copenhagen, Denmark.

2022. *Characteristics and Risk Factors for COVID-19 Outcomes in a Swedish Population-Based Cohort of COPD Patients from the SCIFI-PEARL Project*. Oral presentation, NordicEPI 2022, Reykjavik, Iceland.

2021. *Characteristics and Outcomes of COVID-19 Patients with Pre-Existing COPD in Sweden: A Population-Based Register Study*. Oral presentation, 13th Nordic Pharmacoepidemiological Network (NorPEN) Meeting (Virtual).

Teaching & Mentoring

University of Gothenburg *Gothenburg, Sweden*

Classroom Teaching *2025–2026*

- Register Epidemiology (basic epidemiological concepts) – Autumn 2025, 2026
- Epidemiological Theory and Methods (critical appraisal of scientific papers) – Autumn 2026

University of Gothenburg *Gothenburg, Sweden*

Research and Statistical Supervisor *2023–2024*

- 2024 – Freja Söderbergh (Medical Student), COVID-19 in patients with bronchiectasis.
- 2023 – Omar Noor (Medical Student), COVID-19 in patients with bronchiectasis.
- 2023 – Ágústa Eyjólfsdóttir (Visiting Medical Student, University of Iceland), Predictors of COVID-19 vaccination among immigrants.

Peer Review & Professional Service

Peer Review Activities

Ad hoc Journal Reviewer
Therapeutic Advances in Respiratory Disease (2023–)
Journal of Experimental Pharmacology (2024–)
Respiratory Research (2024–)
CHEST (2024–)
Journal of Diabetes & Metabolic Disorders (2026–)

Other Editorial Roles
Abstract Reviewer, International Society for Pharmacoepidemiology (ISPE) Annual Meeting (Halifax 2023; Berlin 2024; Washington, DC 2025; Milan 2026).

Scientific Societies

International Society for Pharmacoepidemiology (ISPE), Member (2021–Present)
Nordic Pharmacoepidemiologic Network (NorPEN), Member (2022–Present)
BREATHE Sweden Consortium (2022–Present)
Observational Health Data Sciences and Informatics (OHDSI), Member (2023–Present)
– OHDSI Respiratory Working Group (2023–Present)
Swedish Epidemiological Society (SVEP), Member (2024–Present)